EXP NO: 9

DATE: 4-03-2026

DETERMINATION OF Z PARAMETERS OF TWO PORT NETWORK

AIM:

To calculate and verify Z parameters of a two port network using Multisim Simulator.

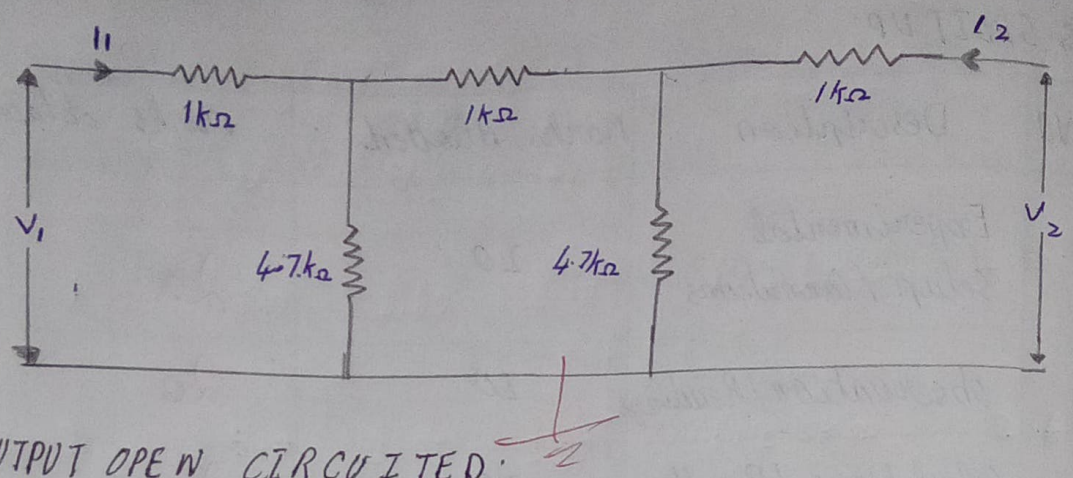
APPARATUS REQUIRED:

S.no	Components	Specification	Quantity
1.	Power Supply	--	1
2.	Multimeter	--	1
3.	Resistors	10kΩ, 10kΩ, 4.7kΩ	
4.	Bread board	--	1
5.	Connecting wires	---	---
6.	PC with Multisim Simulator	---	---

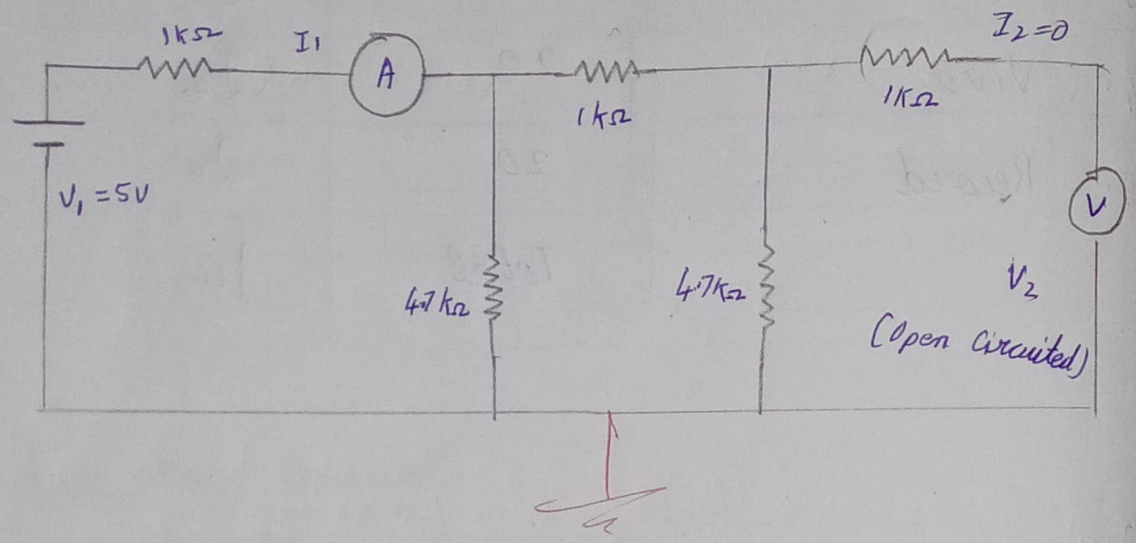
PROCEDURE:

1) Connect the circuit as shown in figure.
2) First open the O/P terminal & supply 5V to I/P terminal.
3) Secondly, open I/P terminal & supply 5V to O/P terminal.
4) Calculate the values of Z parameter using Equation (1) & (2).
5) Switch 'OFF' the supply after taking the readings.

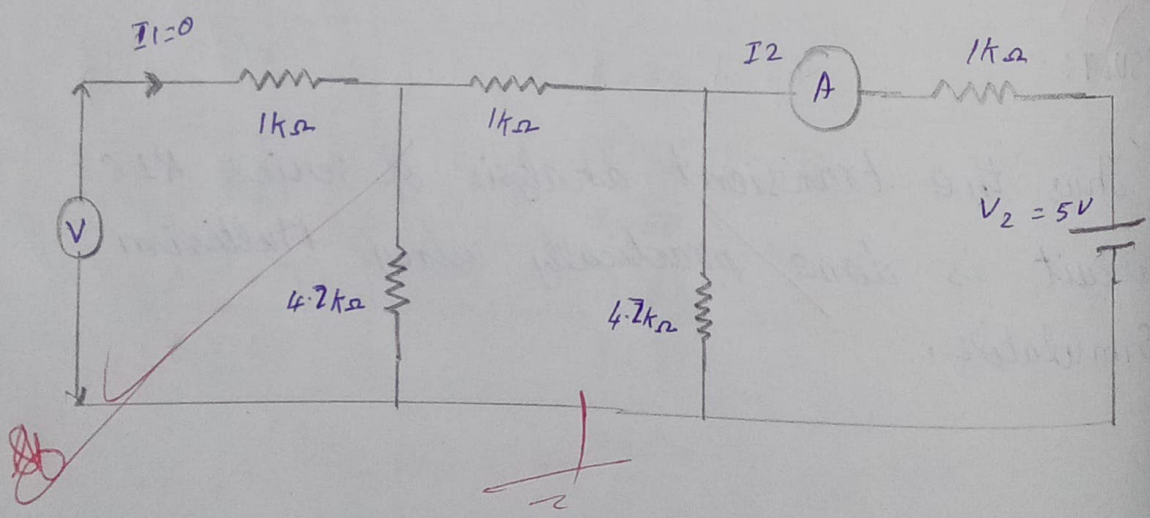
CIRCUIT DIAGRAM:



OUTPUT OPEN CIRCUITED:



INPUT OPED CIRCUITED:



OBSERVATION :

S.No:	OUTPUT OPEN CIRCUITED ($I_2 = 0$)			$Z_{21} = V_2 / I_1 (\Omega)$	$Z_{11} = V_1 / I_1 ($
	V_1	V_2	I_1		
1	5.0	2.60	2.10	1.23 kΩ	2.38 kΩ
2	5.0	2.55	2.08	1.22 kΩ	2.40 kΩ
3	5.0	2.63	2.12	1.24 kΩ	2.35 kΩ

S.NO	INPUT OPEN CIRCUITED ($I_1 = 0$)			$Z_{12} = V_1 / I_2 (\Omega)$	$Z_{22} = V_2 / I_2 ($
	V_2	V_1	I_2		
1	5.00	2.6	2.10	1.23	4.00
2	5.0	2.57	2.09	1.22	2.39
3	5.0	2.62	2.12	1.24	2.38

Verified

MARKS SPLIT-UP:

SI. NO	DESCRIPTION	MARKS ALLOTTED	MARKS OBTAINED
1	EXPERIMENTAL SETUP / CONNECTIONS	20	20
2.	Observation / Readings	20	20
3.	Calculations / Results	20	20
4.	Viva	20	20
5.	Record	20	20
		Total	100

100
17/3

RESULT:

The Z-parameters of the two port network has been calculated and verified by using Multisim Simulator.